Analysis of EEG Signal Evoked by Passive Movement and Motor Imagery

Zhangliang Chen, Qilian Liang and Baoju Zhang

Abstract In recent years, research based on the Motor Imagery (MI) and physical training of stroke rehabilitation therapy has become a hot topic. Among them, brain–computer interface (BCI)-based electrical stimulation (ES) and MI synchronously induced electroencephalo-graph (EEG) analysis is one of the research directions. This paper integrates MI with ES, and introduces the steady-state somatosensory evoked potential (SSSEP) on the MI, and then fuses features. The purpose is to achieve better accuracy, reducing training time, and enhancing target body imagine attention. We use ES and MI synchronous-evoked EEG analysis for rehabilitation training strategy in the process of synchronization. Based on the time–frequency diagram and brain topographic map, we make analysis on event-related desynchronization (ERD), time–frequency characteristic of the steady-state evoked potentials (SSEP), energy characteristics, and distribution regularity of brain regions of each task model. We apply the short-time Fourier transform (STFT), common spatial pattern (CSP), and filter bank common spatial pattern (FBCSP) to classify task pattern recognition on the basis of support vector machine. The results show that based on the lead less the FBCSP has certain superiority, the classification accuracy of ES versus ES&MI achieved 95% or more.

Keywords Stroke · Brain–computer interface · Functional electrical stimulation · Motor imagery · Event-related desynchronization · Common spatial pattern · Support vector machine

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1 Introduction

In recent years, the passive movement and motor imagery (MI) synchronously induced electroencephalo-graph (EEG) analysis has become a commonly used method for the treatment of stroke [1]. However, due to the synchronous process produced by different body motions induced by EEG physiological information and its limitations, the analysis methods of the specific changes of the EEG signals are unclear. Therefore, this paper will discuss the range of low electrical stimulation (does not produce movement of stimulation intensity, ES) exercise imagine EEG change at the same time, to seek the better rehabilitation training strategy in the process of synchronization.

In experiments we use noninvasive BCI system, electrodes cap collected EEG signals, in total of 64 lead lists [2]. The rest of this paper is organized as follows. In Sect. 2, we introduce procedure of experiment. In Sect. 3, we use time–frequency diagram and brain topographic map to analyze EEG signals. Section 4 is characteristics analysis of EEG signals. Section 5 is simulations of classification accuracy and Sect. 6 concludes this paper.

2 Experimental Scheme

In order to guarantee the success rate of experiments, experimenters are all trained. We collect five volunteer subjects of EEG signals. It includes three men and two women whom aged between 22 and 25 and all of them are right-handed. The whole experiment includes four paradigms. In the first paradigm we ask volunteers to image their right-hand clenched fists action and collect its EEG. In the second paradigm volunteers not only image fisting action but also are added ES on their right hands at the same time, and we collect their EEG. In the third paradigm we add ES on volunteers right hands only and collect their EEG. The fourth paradigm is a rest paradigm which means volunteers can relax and we collect their EEG. Each paradigm has 40 single repeat in 10 s, and the whole experiment has a total of 160 EEG data samples. Task model of four paradigms is same and shown in Fig. 1, divided into four periods, a total of 10 s.

First period is preparation period. There is a stationary white circle at the central of computer screen and lasting 2 s. It means experiment start, and volunteers are keeping relaxed state of no action. Second period is prompt period. In this period the white circle disappears and a red circle appears. Lasting 2 s which means prompting volunteers to prepare since imagine action thinking task is about to begin. For the second and the third paradigms, we give volunteers right hands ES at the same time. Third period is the imagination period. In this period red circle disappears but a right arrow appears and lasting 4 s. This signal asks volunteers to

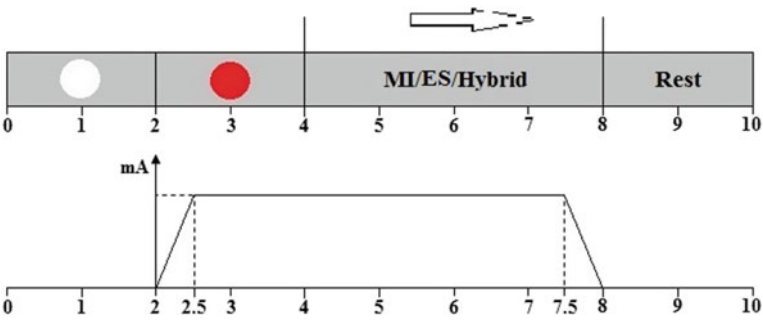


Fig. 1 Task model diagram

focus on their right hands ES and imagine the action of their right hands clench fists. The fourth period is for recovery with the arrow disappears. A stationary word REST appears. Volunteers keep resting state [3]. Each paradigm of MI experiments includes 40 times single task and a total of 40 * 10 s = 400 s.

3 Motor Imagery Characteristic Analysis

3.1 Time–Frequency Diagram

Figure 2 shows the average time–frequency diagram of experimental stage of the six volunteers in the C3, CZ, C4 lead lists in four experimental models. The horizontal axis is time and the vertical axis is frequency. It can be seen from the figure that the brain electrical signal evoked by the MI shows obvious event-related desynchronization (ERD) features and the obvious steady-state somatosensory

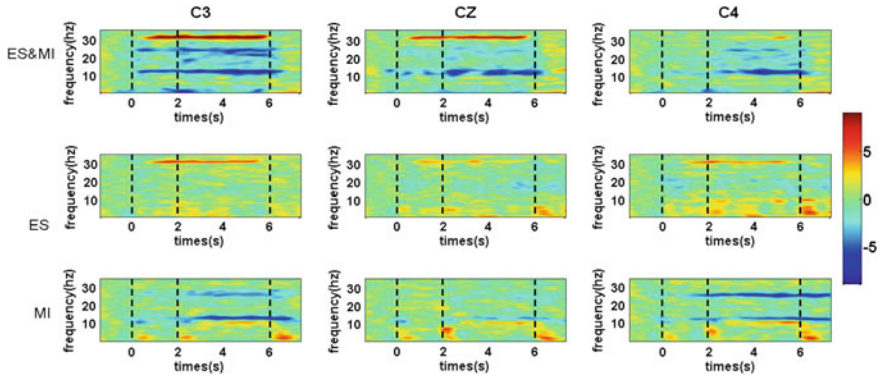


Fig. 2 C3 C4 CZ lead lists time–frequency diagram; blue represents the energy drops; red represents the energy rise

evoked potential (SSSEP) characteristics of EEG signals evoked by the external electrical stimulation in the frequency domain [4, 5]. At the beginning of the 4 s MI, some significant energy attenuation phenomena occur in the 0.5 s, which appear in the alpha band (8–9 Hz) and the beta band (15–17 Hz). The ERD characteristic of alpha frequency band is more obvious and lasts about 3S time than the beta frequency band. When the volunteers are in a resting state, no ERD or SSSEP features are in the EEG.

It can be seen from the diagram that when volunteers imagine the right fist action the ERD characteristics of the corresponding C3 lead list is very obvious. CZ lead list which reflects adjacent brain regions reaction also appears ERD phenomenon but is not so obvious at lead list C3. When the volunteers imagine right fist action at the same time is added outside ES, C3 lead list appears SSSEP phenomenon at 31 Hz in addition to strong external ERD characteristics.

3.2 Brain Topographic Map

Alpha band ERD phenomenon has the characteristics of good duration and intensity as well as more conducive to the follow-up observation and analysis, so we use alpha band ERD phenomenon for the main analysis.

We use window function to intercept ERD and SSSEP characteristics at obvious band of time–frequency diagram in order to select the best data for making brain topographic maps. We get ERD and SSSEP single-dimensional feature data of time–frequency within the window, and average each lead list separate time–frequency superimposed data, gaining feature data for brain map needed. Using the above method we make MI only, ES only, and ES&MI brain topographic maps of ERD and SSSEP characteristics as shown in Fig. 3.

From the brain topographic maps seen in the MI and ES experimental paradigms, it is obvious that the left side of the brain power attenuation is more obvious than the right side, with the obvious ERD and SSSEP contralateral dominant phenomenon [6].

4 MI Feature Extraction Based on Common Spatial Pattern

In this section, we use filter bank common spatial pattern (FBCSP) to analyze the characteristics of EEG. After EEG signal acquisition the whole 8–32 Hz is divided into six bands. The common spatial pattern (CSP) is used to extract the features and add the 1 Hz frequency doubling [7]. Then we use the feature selection algorithm to automatically select the recognition force of the dual band CSP and the corresponding characteristics.

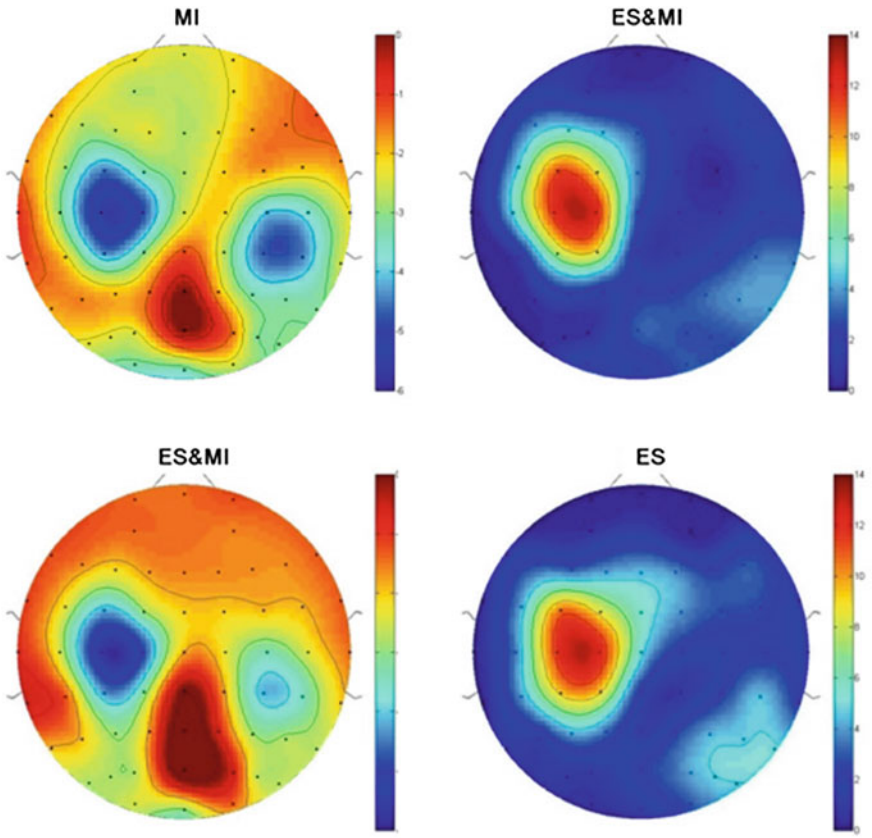


Fig. 3 ERD and SSSEP brain topographic maps of three experimental paradigms

$$X_{CSP} = W^T * X, \tag{1}$$

where X_{CSP} is obtained from original EEG signal X after filtering, W is the filter matrix, each column vector $w_j \in W^{N\times N}(j = 1, \ldots, N)$ is a filter, and $A = (W^{-1})^T$ is spatial pattern matrix, where each column $a_j \in A^{N\times N}(j = 1, \ldots, N)$ is a vector space model.

Assuming that the first-class matrix is represented by X_l, the second-class matrix is expressed as X_r, and the dimension is the lead number * sampling number. The spatial covariance matrix can be expressed as

$$R_l = \frac{X_l X_l^T}{trace(X_l X_l^T)} \qquad R_r = \frac{X_r X_r^T}{trace(X_r X_r^T)}, \tag{2}$$

where T denotes transpose, and trace *trace(X)* is the sum of diagonal elements.

X_l and X_r can be linearly expressed as

$$X_l = [C_l C_c] \begin{bmatrix} S_l \\ S_c \end{bmatrix} \qquad X_r = [C_r C_c] \begin{bmatrix} S_r \\ S_c \end{bmatrix}, \tag{3}$$

where S_l is the signal source for the first class, S_r is the signal source for the second kinds of needles, and C_l and C_r are the corresponding feature vectors.

The purpose is to establish a spatial filters F_l and F_r, which can be used to extract the components of the signal sources S_l and S_r:

$$S_l = F_l * X_l \qquad S_r = F_r * X_r, \tag{4}$$

where F_l and F_r are the first and second types of action corresponding to the spatial filter, through the use of data training.

R_l and $\mathrm{R_r}$ are the mean of covariance normalization::

$$\overline{R_l} = \frac{1}{N_l} \sum_{i=1}^{N_l} R_l(i) \qquad \overline{R_r} = \frac{1}{N_r} \sum_{i=1}^{N_r} R_r(i), \tag{5}$$

where N is the number of experiments in the training data.

Then use the principal component analysis

$$\overline{R} = \overline{R_l} + \overline{R_r} = U_0 \Lambda U_0^T, \tag{6}$$

where U_0 is a feature vector and Λ is the characteristic value.

Thus the whitening matrix is expressed as

$$P = \Lambda^{1/2} U_0^T. \tag{7}$$

Thus can be expressed as

$$Y_l = P \overline{R_l} P^T \qquad Y_r = P \overline{R_r} P^T. \tag{8}$$

Based on principal component analysis,

$$Y_l = U \Lambda_l U^T \qquad Y_r = U \Lambda_r U^T. \tag{9}$$

It can be shown that the eigenvalues of Y_l and Y_r have diagonal matrix $\Lambda_B = I$, where I is the unitary matrix. So the final corresponding to the spatial filter F_l and Fr can be expressed as

$$F_l = U_l^T P \qquad F_r = U_r^T P. \tag{10}$$

5 Simulation

In this section, we use support vector machine (SVM) to classify feature vector of each training sample, and then draw the correct classification rate [8].

At the beginning we use all lead lists EEG data to classify feature vectors: the classification of ES versus REST as a result of the control group and the classification of ES&MI versus ES as the experimental group. Comparing these two results as shown in Table 1, we obtain more than 95% accuracy rate in ES&MI versus ES, which means our purpose is achieved. However, taking into account the full lead lists lead to patients discomfort, we use nine lead lists (C3, C4, CZ and their up and down lists) for classification as shown in Table 2.

We can see the accuracy rate of the classification of the nine lead lists. ES&MI versus ES has been greatly reduced. In order to improve the accuracy rate we use fast Fourier transform (FFT) to process EEG signal. After that we add amplitude at 31 Hz and frequency doubling at 62 Hz for feature extraction and classification. Each bandwidth is 1 Hz. Table 3 shows the classify accuracy rate.

Although the correct rate has improved, it is still lower than full lead lists. So finally on the basis of FFT we use FBCSP for feature extraction and classification. The accuracy is summarized in Table 4.

At this time the accuracy rate has reached more than 95%, infinitely close to the accuracy rate of the full lead lists.

Table 1 CSP all lead lists feature extraction and classification accuracy rate

CSP all lists classification accuracy rate	MI versus REST (%)	ES&MI versus ES (%)
Volunteer 1	100	97.5
Volunteer 2	98.75	100
Volunteer 3	100	100
Volunteer 4	100	100
Volunteer 5	100	80
Average	99.75	95.5

Table 2 CSP 9 lead lists feature extraction and classification accuracy rate

CSP 9 lists classification accuracy rate	MI versus REST (%)	ES&MI versus ES (%)
Volunteer 1	98.75	87.5
Volunteer 2	88.75	87.5
Volunteer 3	100	98.75
Volunteer 4	98.75	93.75
Volunteer 5	98.75	68.75
Average	97	87.25

Table 3 CSP+FFT 9 lead lists feature extraction and classification accuracy rate

CSP+FFT 9 lists classification accuracy rate	MI versus REST (%)	ES&MI versus ES (%)
Volunteer 1	98.75	90
Volunteer 2	88.75	93.75
Volunteer 3	100	96.25
Volunteer 4	98.75	95
Volunteer 5	98.75	83.75
Average	97	91.75

Table 4 FBCSP+FFT 9 lead lists feature extraction and classification accuracy rate

FBCSP+FFT 9 lists classification accuracy rate	MI versus REST (%)	ES&MI versus ES (%)
Volunteer 1	100	93.75
Volunteer 2	91.25	98.75
Volunteer 3	100	100
Volunteer 4	98.75	95
Volunteer 5	100	88.75
Average	98	95.25

6 Conclusions

First, we designed the movement pattern of MI, and the experimental type of ES was combined with MI, which constituted the four kinds of task paradigms. Second, we performed spatial filtering and denoising data preprocessing for the original EEG data. Third, we made qualitative analysis on the phenomenon of SSSEP induced by ES and the phenomenon of ERD induced by MI. Fourth, we used the CSP and FBCSP two classifications of spatial pattern algorithm add FFT algorithm. On the basis of that we combined CSP+FFT and FBCSP+FFT, two better feature extraction algorithms. Finally, we used SVM pattern recognition method to classify EEG data from fourtypes of tasks and compared the classification accuracy of MI versus REST with ES&MI versus ES. The classification accuracy is more than 95%. So we can conclude that MI&ES which can provide high classification accuracy on the basis of good patient comforts and convenient experiments is a better rehabilitation method in stroke rehabilitation strategy.

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